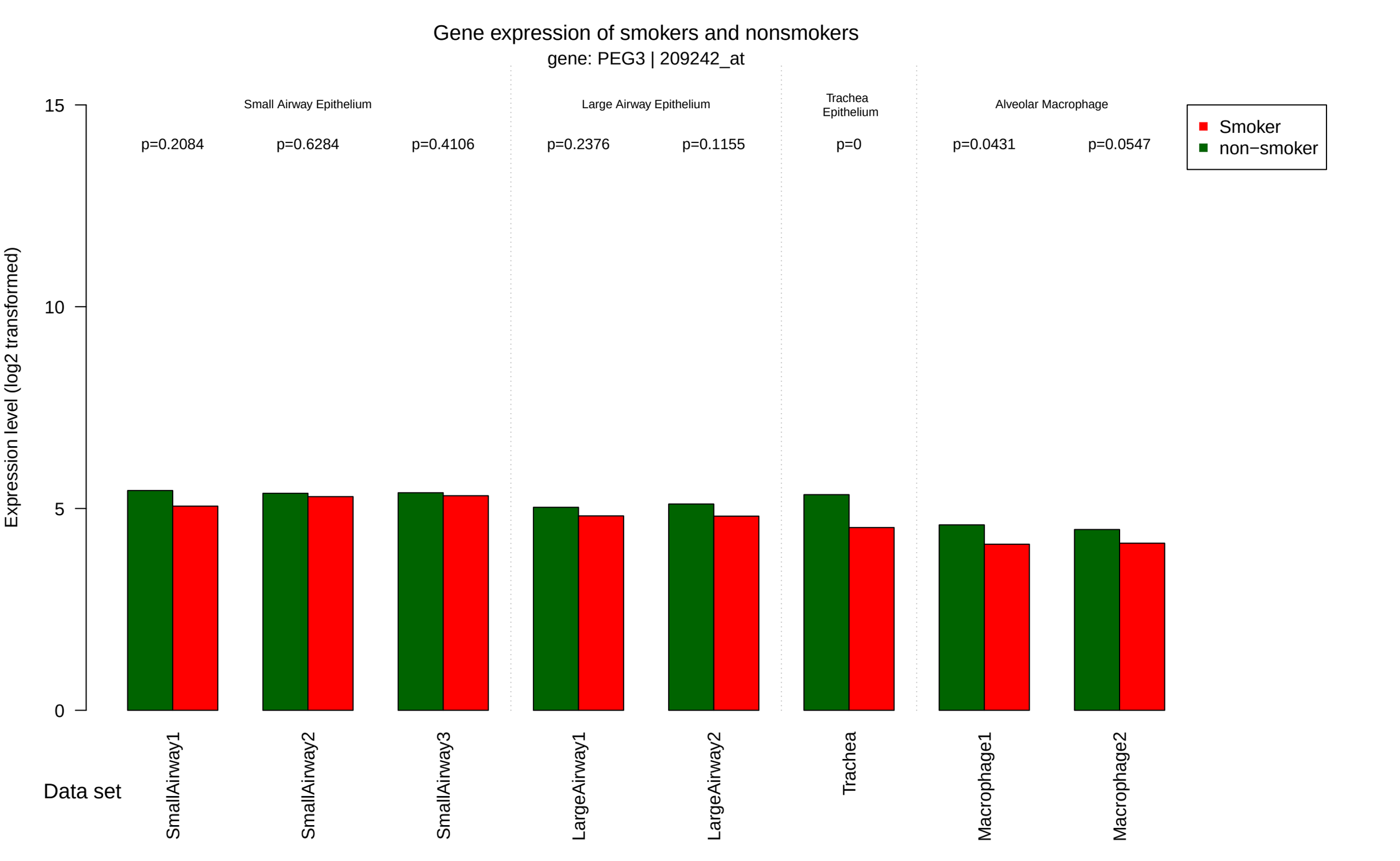
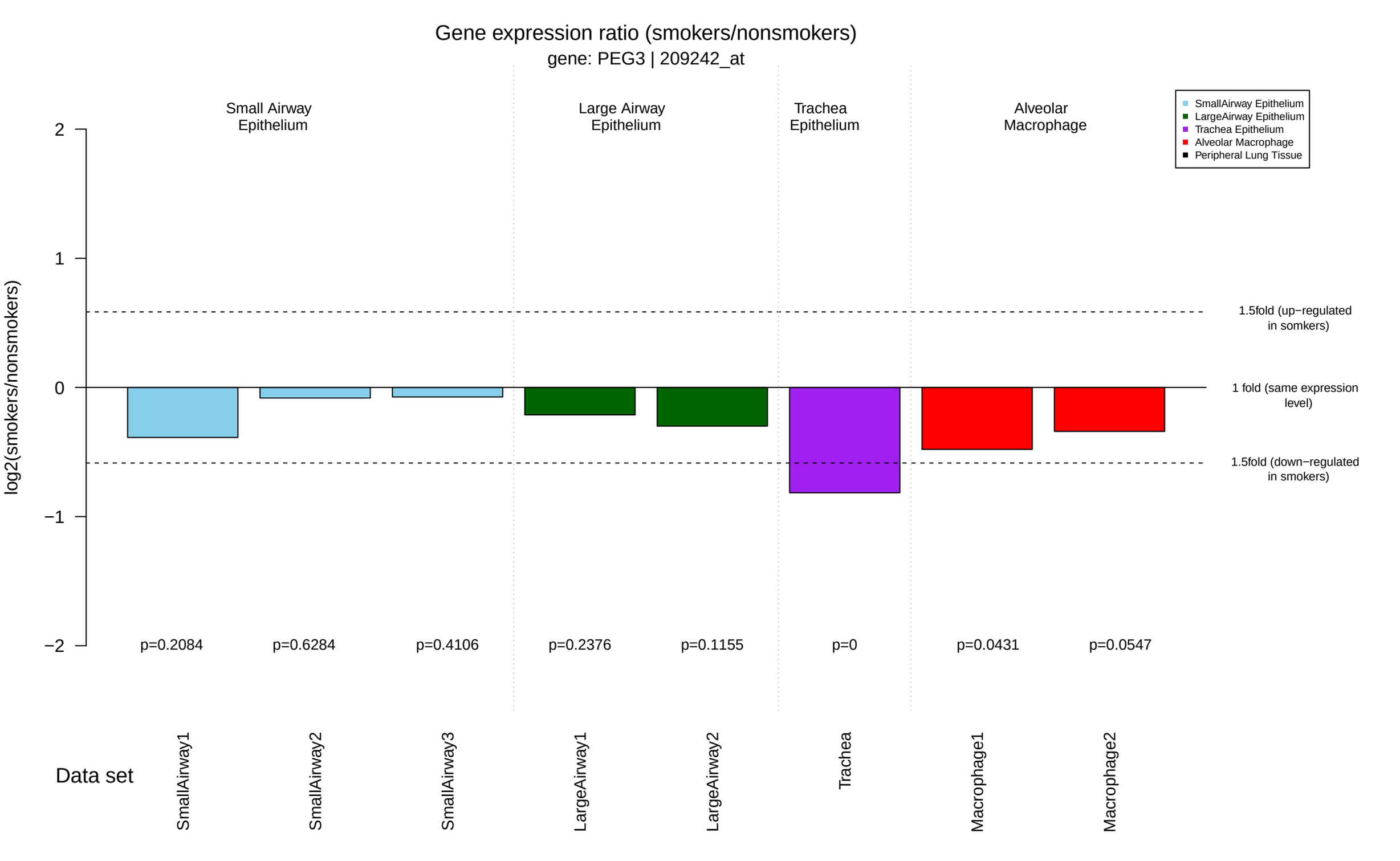
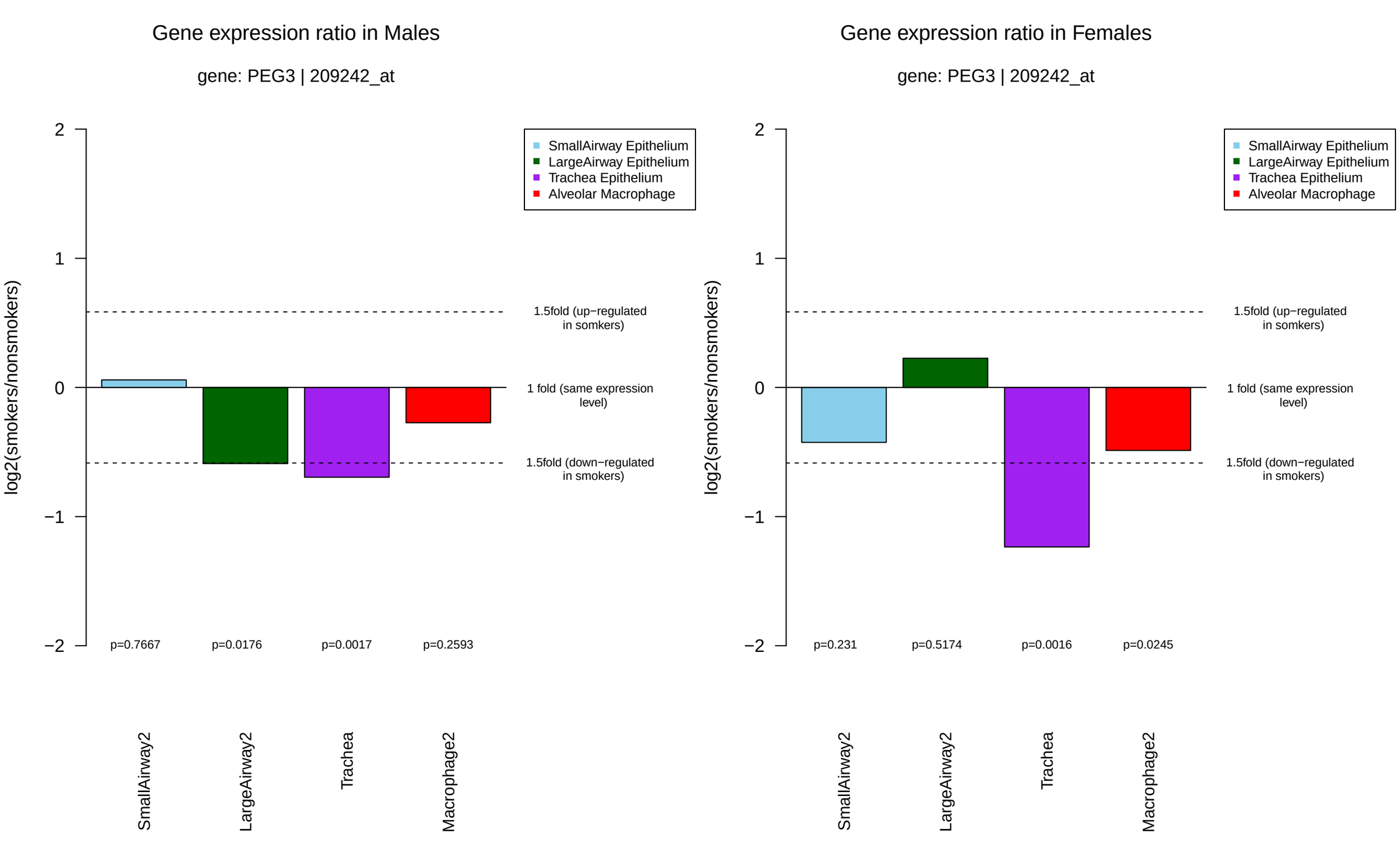
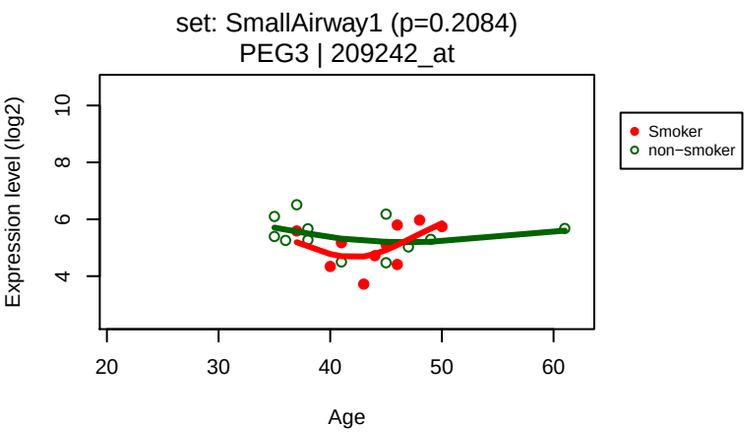
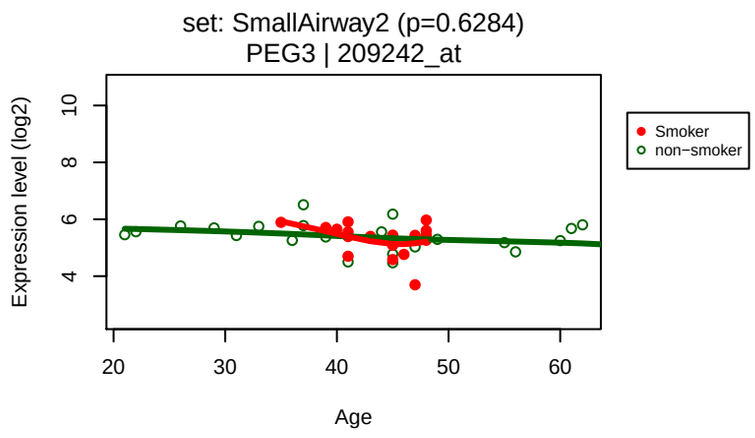
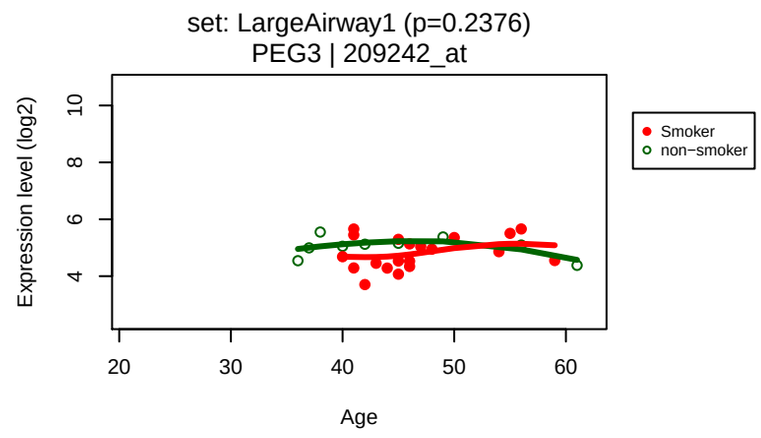

Gene expression of smokers and nonsmokers
gene: PEG3 | 209242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2084
p=0.6284
p=0.4106
p=0.2376
p=0.1155
p=0
p=0.0431
p=0.0547
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

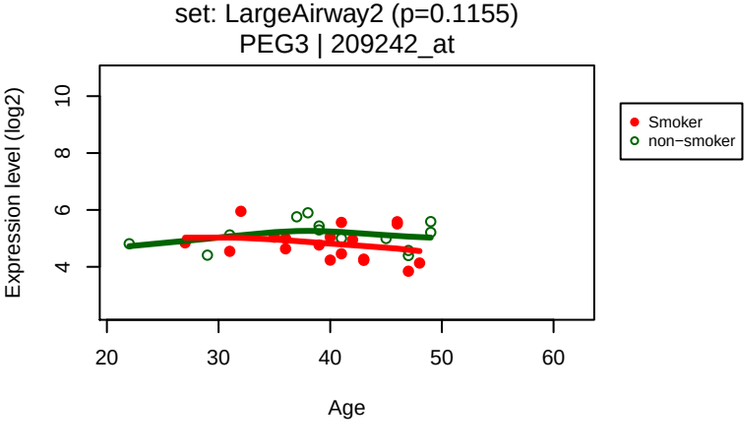

Gene expression ratio (smokers/nonsmokers)
gene: PEG3 | 209242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2084
p=0.6284
p=0.4106
p=0.2376
p=0.1155
p=0
p=0.0431
p=0.0547
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

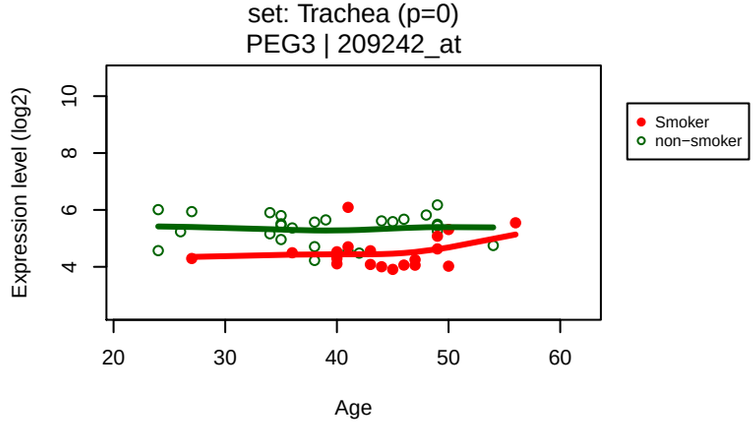

Gene expression ratio in Males
gene: PEG3 | 209242_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7667
p=0.0176
p=0.0017
p=0.2593
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PEG3 | 209242_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.231
p=0.5174
p=0.0016
p=0.0245
SmallAirway2
LargeAirway2
Trachea
Macrophage2

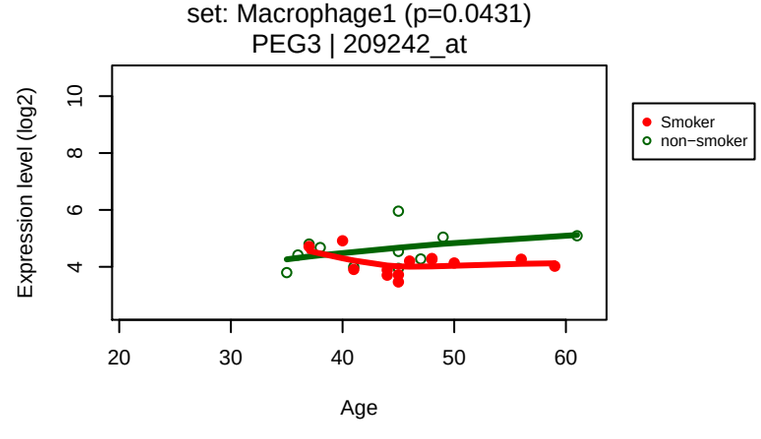

set: SmallAirway1 (p=0.2084)
PEG3 | 209242_at
Expression level (log2)
Age
Smoker
non−smoker

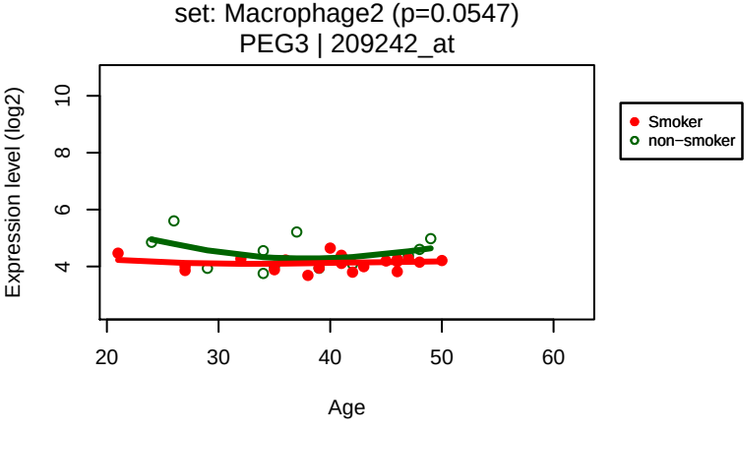

set: SmallAirway2 (p=0.6284)
PEG3 | 209242_at
Expression level (log2)
Age
Smoker
non−smoker

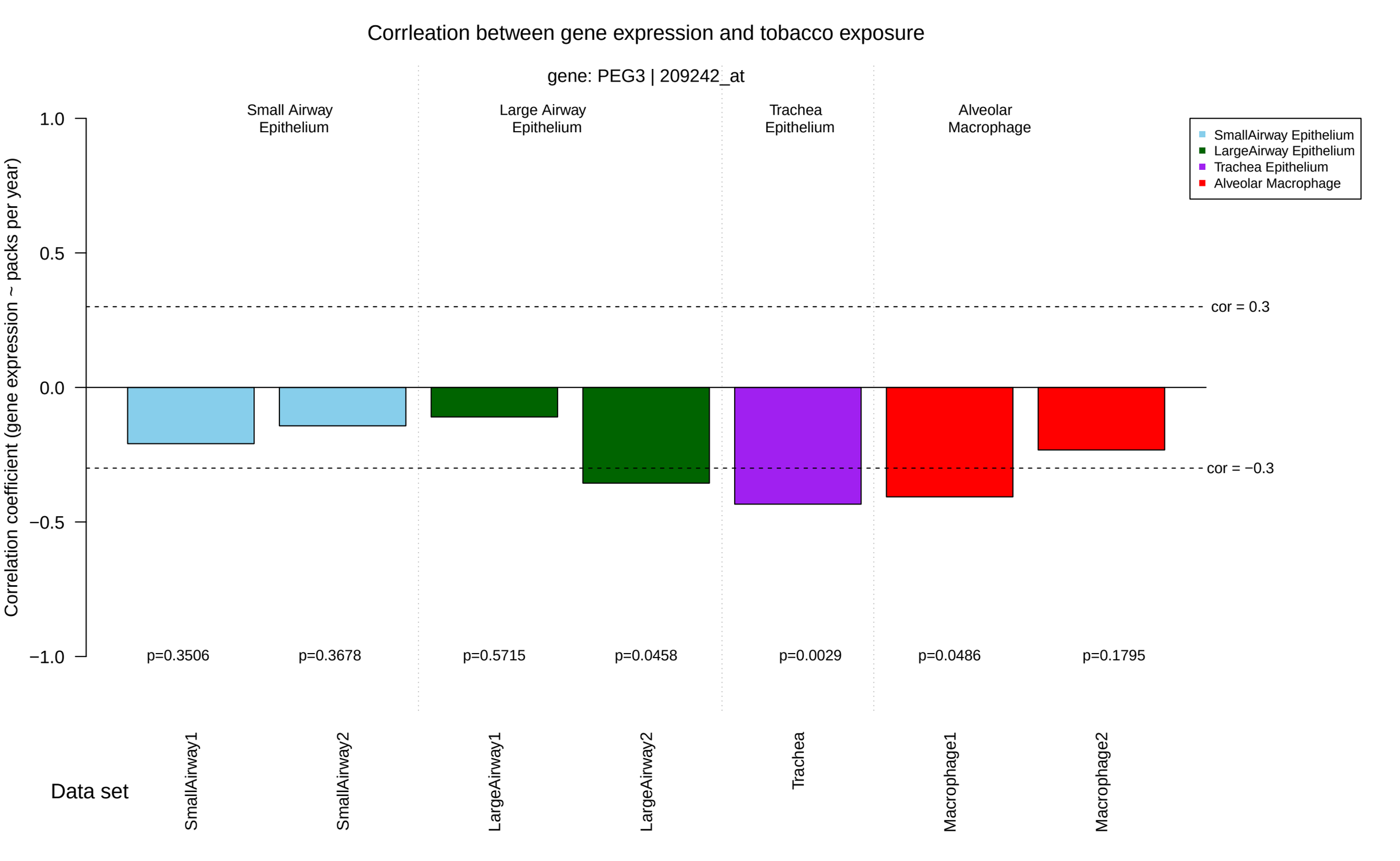

set: LargeAirway1 (p=0.2376)
PEG3 | 209242_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1155)
PEG3 | 209242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
PEG3 | 209242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0431)
PEG3 | 209242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0547)
PEG3 | 209242_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PEG3 | 209242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3506
p=0.3678
p=0.5715
p=0.0458
p=0.0029
p=0.0486
p=0.1795
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2